Lecture 14
Statistical Significance and an Introduction to Probability

Review

1 What is a sampling distribution?

The distribution of a statistic over all possible samples of the same size n.

2 What is the standard error?

The standard deviation of a sampling distribution; for $\bar{x}$ it is $SE = \sigma / \sqrt{n}$.

3 What is the margin of error?

How far an estimate is expected to fall from the parameter: $2 \times SE$.

4 What is a linear transformation?

$y = ax + b$: it rescales and shifts the center and spread, but never changes the shape.

What you should know: the normal distribution

The two parameters

$x \sim N(\mu, \sigma)$: μ locates the center, σ sets the spread.

The empirical rule

68%, 95% and 99.7% within 1, 2 and 3 standard deviations (bell-shaped only).

The z-score

$z = (x - \bar{x}) / s$ counts standard deviations from the mean; $|z| \geq 2$ flags an outlier.

Data: student heights

A survey of 379 college students recorded each student's height and gender.

X = the height of a student, in inches

HEIGHT	GENDER
56	Female
57	Female
58	Female
60	Female
60	Female
…	…

n = 379 students (the file lists the women first, then the men, each sorted by height)

Statistical significance

Someone hands you 20 students and says they are a random sample of all 379.

$\bar{x}$ = 70.33 in μ = 67.1 in SE = 0.92 in

Is this result statistically significant?

z = (70.33 – 67.1) / 0.92 ≈ 3.5: far outside the margin of error, so yes
Statistically significant: this was not a random sample of all students (they were all men)

Why probability?

Wednesday: about 95% of samples land within 2 SE of μ.

Today: a sample that far out would be very unlikely to happen by chance.

We only ever take one sample. So what do "95%" and "unlikely" actually mean?

They describe what would happen if we repeated the random sampling many, many times

Statements about the long run of a chance process are statements of probability

Quantifying uncertainty

randomization

we use chance on purpose - random sampling, random assignment - to avoid bias

the price: uncertainty

by chance, the statistics from a sample are not exactly the parameters of the population

the payoff

chance-based results follow the laws of probability, so we can measure how far off we may be

The language of probability

random trial

a process or experiment with well-defined possible outcomes; it is random because more than one outcome is possible

sample space S

the collection of all possible outcomes: for one roll of a die, S = {1, 2, 3, 4, 5, 6}

event

any possible outcome of a random trial, written with a capital letter: A = roll an even number = {2, 4, 6}

probability P(A)

a number between 0 and 1 that measures how likely event A is: the proportion of times A occurs in a very long series of repetitions

Equally likely outcomes

Roll a fair six-sided die: all six outcomes are equally likely.

P(A) = (number of ways A can happen) / (total number of outcomes)

(a) What is P(even)?
(b) What is P(not a six)?

(a) 3 / 6 = 0.5
(b) 5 / 6 ≈ 0.83

Example: probabilities from real data

Pick one of the 379 students from the heights data at random.

	At least 70 in	Under 70 in	Total
Female	19	243	262
Male	87	30	117
Total	106	273	379

(a) What is P(the student is at least 70 in)?

(b) What is P(the student is female)?

(a) 106 / 379 ≈ 0.28

(b) 262 / 379 ≈ 0.69

Where do probabilities come from?

Flip a fair coin once. Intuition says P(heads) = 1/2.

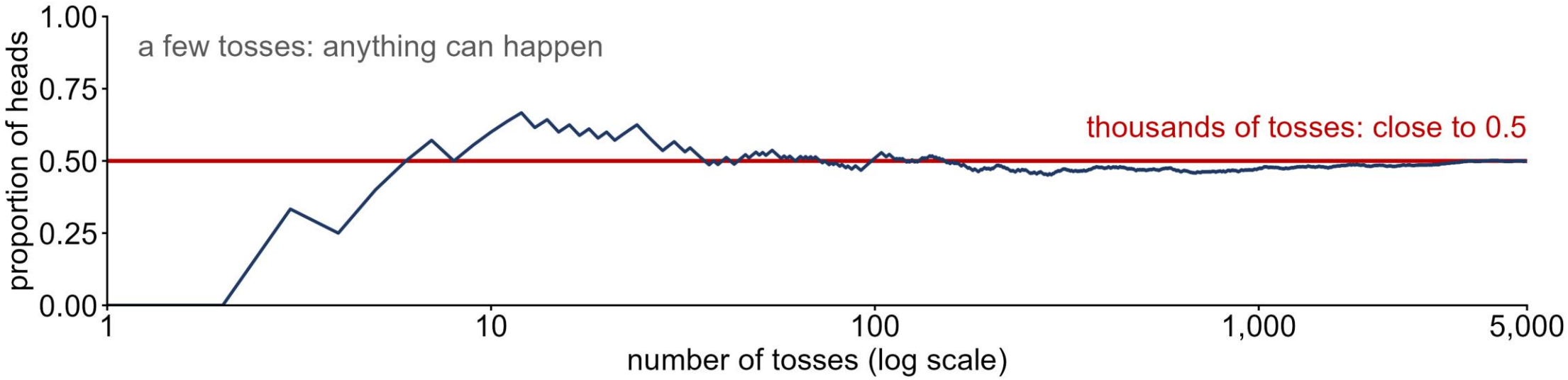
You flip it 4 times and get 3 heads: 75%. Was 1/2 wrong?

No: over a few flips the proportion is unpredictable; over many it is very predictable
Probability: the proportion of times an event occurs in a very long series of repetitions

The law of large numbers

law of large numbers

as the number of independent trials n increases, the relative frequency of an event, (number of times A occurs) / n, gets closer and closer to P(A)



Independent trials: the outcome of one trial does not affect the outcome of any other
Jacob Bernoulli proved it (published 1713); Karl Pearson tossed a coin 24,000 times: 12,012 heads = 0.5005

The rules of probability

Rule 1: $0 \leq P(A) \leq 1$

the probability of an event must be a value between 0 and 1

Rule 2: $\Sigma P(A_i) = 1$

the probabilities of all the outcomes in the sample space must add to 1

Rule 3: $P(A \cup B) = P(A) + P(B)$

the addition rule for DISJOINT events: A and B have no outcomes in common

Rule 4: $P(A') = 1 - P(A)$

the complement rule: A' ("A-not") is every outcome in S that is not in A

Combining two events

union $A \cup B$

outcomes in A or in B (or both); key word "or"

intersection $A \cap B$

outcomes A and B have in common; key word "and"

disjoint (mutually exclusive)

A and B have no outcomes in common: $A \cap B = \emptyset$

